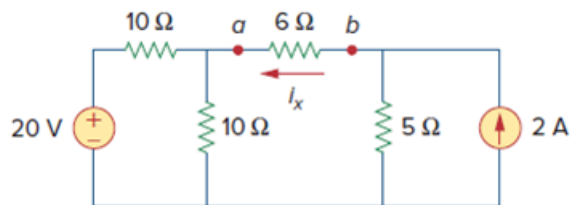


Problem

Find the Thevenin equivalent looking into terminals a - b of the circuit in Fig. 4.110 and solve for i_x .

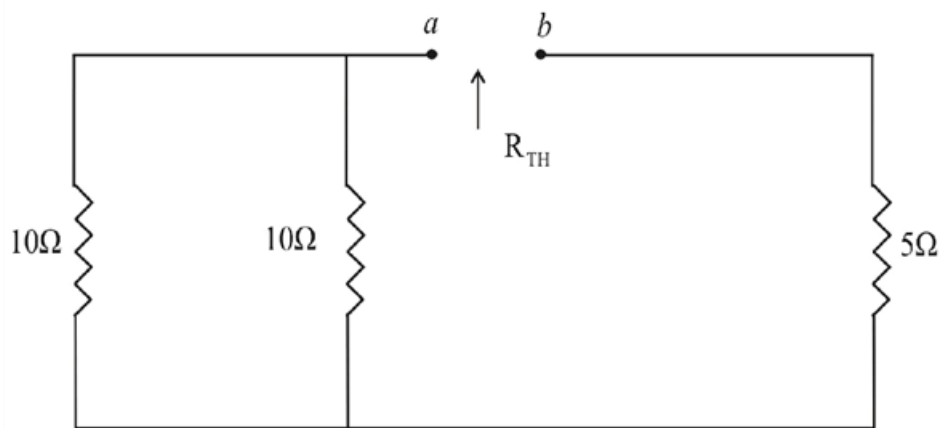
Figure. 4.110:



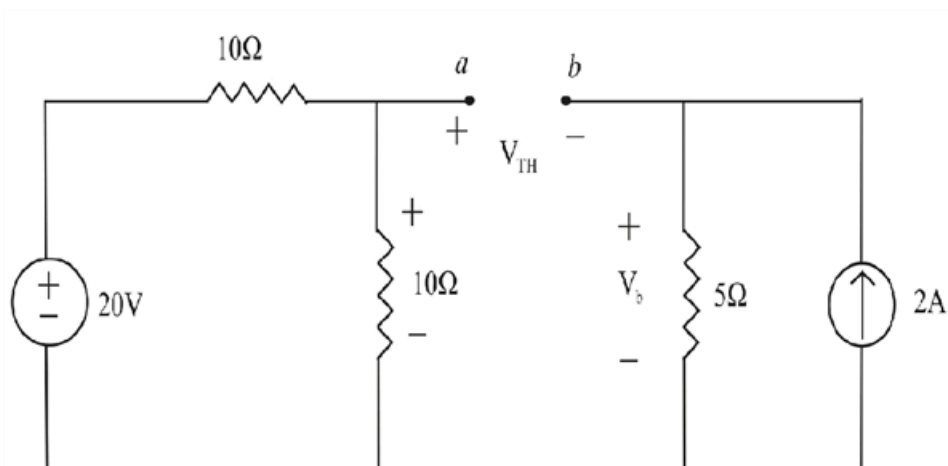
Step-by-step solution

Step 1 of 3

To find R_{TH} , consider the circuit in fig (a).



Step 2 of 3



Step 3 of 3

From the first circuit Thevenin's equivalent resistance is

$$R_{TH} = (10 \parallel 10) + 5$$

$$R_{TH} = 5 + 5$$

$$R_{TH} = 10\Omega$$

To find V_{TH} , Consider the circuit in fig(b).

$$V_b = 2 \times 5$$

$$V_b = 10\text{ V}$$

$$V_a = 20 \left(\frac{10}{10+10} \right)$$

$$V_a = 10\text{ V}$$

From the circuit we can write as

$$-V_a + V_{TH} + V_b = 0$$

$$V_{TH} = 0\text{ V}$$

Therefore Thevenin's voltage 0, so the current $i_x = 0\text{ A}$